

MILESTONE 3

Clean Diversity Analytics Module

Implementation & Technical Documentation

| ITEM                         | DETAIL                                                                                    |
|------------------------------|-------------------------------------------------------------------------------------------|
| Assigned Area                | Diversity Analytics Module                                                                |
| Task Focus                   | Analyze gender, age and demographic metrics; create charts and comparison visualizations. |
| Dataset                      | Uploaded workforce Excel dataset                                                          |
| Original Records             | 1,470                                                                                     |
| Duplicate Records Found      | 0                                                                                         |
| Records After Cleaning       | 1,470                                                                                     |
| Missing Cells After Cleaning | 0                                                                                         |

## 1. Milestone 3 Assigned Task

The assigned Milestone 3 responsibility is the **Diversity Analytics Module**. The implementation covers the two specified activities:

- **Analyze gender, age, and demographic metrics.**
- **Create charts and comparison visualizations.**

The module converts the workforce dataset into structured diversity metrics and visual outputs suitable for integration with the workforce intelligence dashboard.

## 2. Dataset Used

The implementation uses the workforce Excel dataset uploaded for this task. The dataset was processed before the diversity analysis.

| Validation                    | Result |
|-------------------------------|--------|
| Original records              | 1,470  |
| Exact duplicate rows detected | 0      |
| Rows after duplicate removal  | 1,470  |
| Missing cells after cleaning  | 0      |

**Dataset status:** The dataset used for the Milestone 3 analysis is clean with no exact duplicate rows and no missing cells after the cleaning process.

## 3. Data Cleaning & Preparation

The implementation performs a basic repeatable cleaning step before generating the diversity outputs.

- Load the uploaded Excel workforce dataset.
- Trim leading and trailing whitespace from text fields.
- Detect and remove exact duplicate rows.
- Validate the required Gender, Age and Department fields.
- Convert Age to numeric format.
- Create age groups for workforce comparison.
- Use the resulting clean dataset for all diversity calculations and charts.

The cleaning step is applied before analysis so that the visualizations are generated from the prepared workforce dataset.

## 4. Overall Implementation Workflow

**Uploaded Excel Dataset → Data Cleaning → Data Validation → Diversity Metrics → Gender / Age / Demographic Analysis → Comparison Analysis → Charts → Excel Dashboard Outputs**

The implementation produces both visualization files and structured Excel summaries. A reusable Python script is also included in the implementation package.

## 5. Gender Analytics

The Gender Analytics component calculates employee counts and workforce percentages for each gender category. It also creates a department-level gender comparison.

| Output               | Purpose                                            |
|----------------------|----------------------------------------------------|
| Gender Distribution  | Shows the number of employees by gender.           |
| Gender Percentage    | Shows the percentage contribution of each gender.  |
| Gender by Department | Compares gender representation across departments. |



Figure 1. Workforce Gender Distribution

## 6. Age Analytics

The Age Analytics component analyzes the workforce using employee age and predefined age groups.

| Age Group | Range         |
|-----------|---------------|
| 18–25     | 18 through 25 |
| 26–35     | 26 through 35 |
| 36–45     | 36 through 45 |
| 46–55     | 46 through 55 |
| 56+       | 56 and above  |

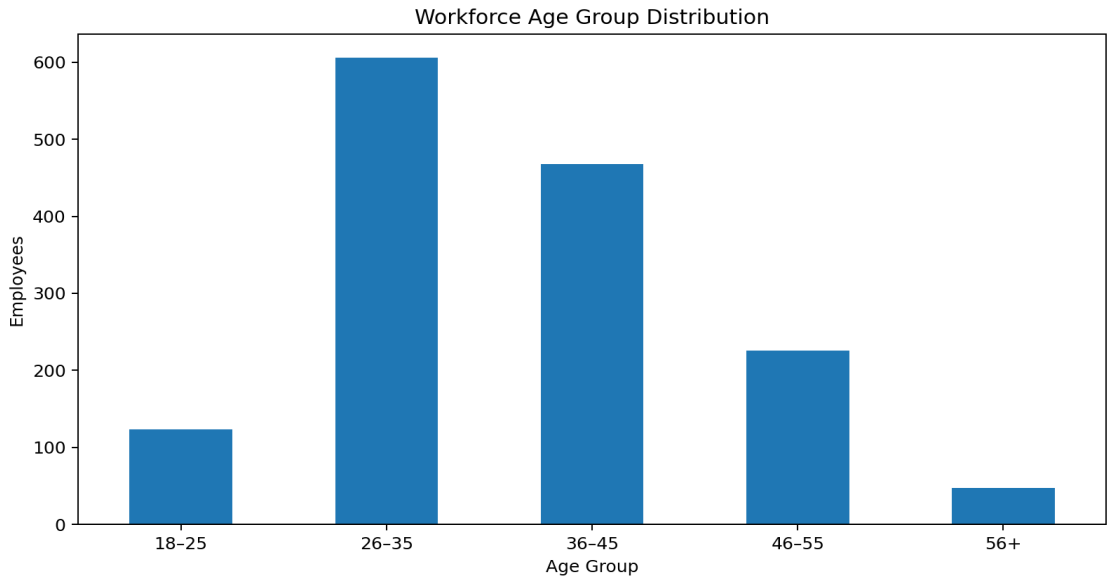


Figure 2. Workforce Age Group Distribution

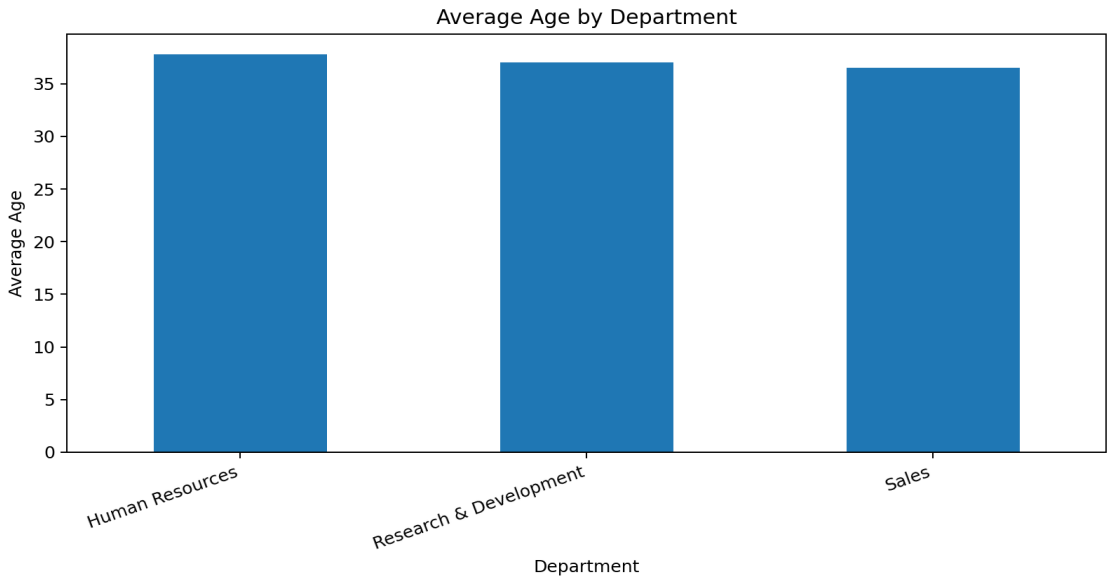


Figure 3. Average Age by Department

## 7. Demographic Analytics

The demographic component uses the available workforce fields to create distribution summaries. The implementation includes the following demographic dimensions where present in the uploaded dataset:

- Department
- Marital Status
- Education
- Job Role
- Education Field
- Business Travel

Each supported dimension is summarized using employee counts and percentages and is exported to the dashboard-ready Excel workbook.

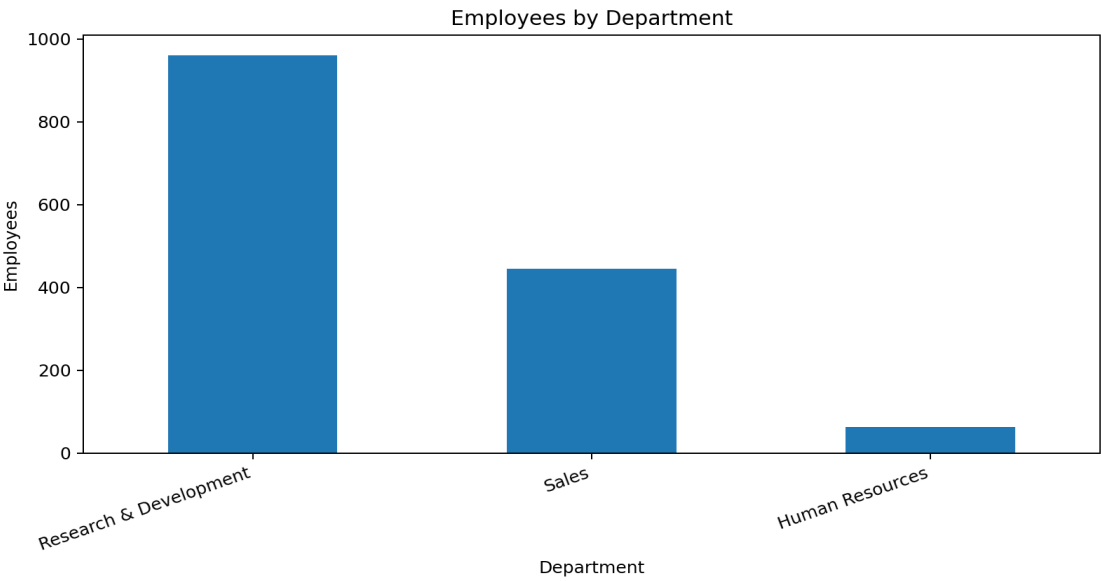


Figure 4. Employees by Department

## 8. Comparison Visualizations

Comparison visualizations provide cross-category analysis required by the Milestone 3 task.

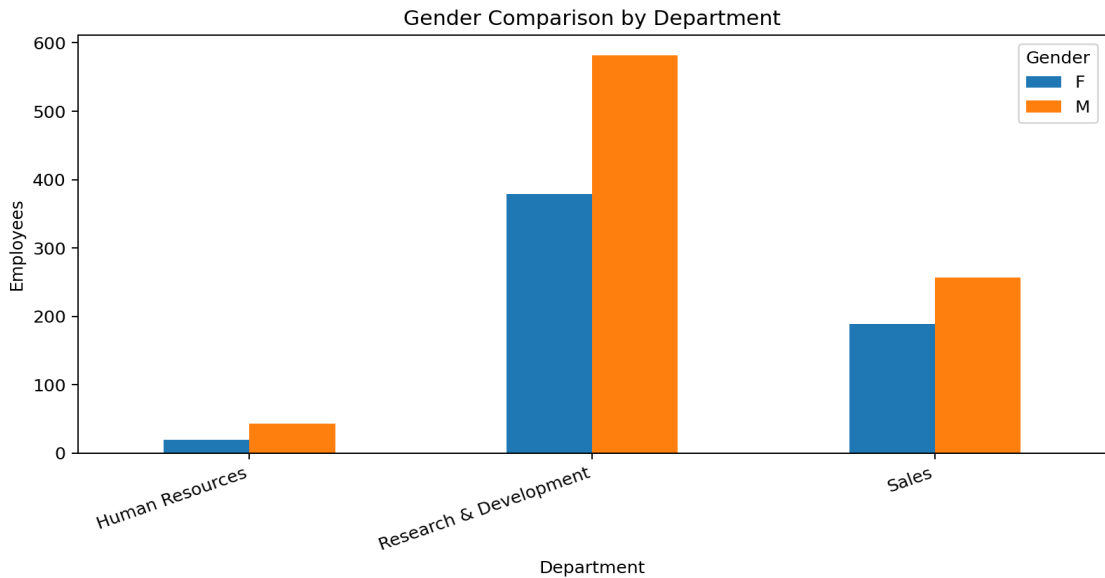


Figure 5. Gender Comparison by Department

The gender comparison chart allows the workforce composition of each department to be viewed side by side. The average-age comparison provides a corresponding age-based comparison across departments.

## 9. Additional Visualizations

| Visualization             | Purpose                                    |
|---------------------------|--------------------------------------------|
| Gender Distribution       | Overall workforce gender composition.      |
| Gender Percentage         | Percentage representation by gender.       |
| Age Group Distribution    | Workforce distribution by age band.        |
| Age Distribution          | Overall employee age distribution.         |
| Department Distribution   | Workforce distribution across departments. |
| Marital Status            | Marital-status composition.                |
| Education Level           | Education-level composition.               |
| Job Role                  | Job-role composition.                      |
| Education Field           | Education-field composition.               |
| Gender by Department      | Department-level gender comparison.        |
| Average Age by Department | Department-level age comparison.           |

## 10. Dashboard-Ready Outputs

The implementation generates an Excel workbook containing structured tables that can be consumed by the workforce intelligence dashboard.

| Sheet             | Content                          |
|-------------------|----------------------------------|
| Gender            | Gender count and percentage.     |
| Age_Groups        | Age-group count and percentage.  |
| Department        | Department count and percentage. |
| Marital_Status    | Marital-status summary.          |
| Education         | Education-level summary.         |
| Job_Role          | Job-role summary.                |
| Education_Field   | Education-field summary.         |
| Business_Travel   | Business-travel summary.         |
| Gender_Department | Gender comparison by department. |
| Age_Department    | Age statistics by department.    |

## 11. Implementation Files

| File / Folder                    | Purpose                                            |
|----------------------------------|----------------------------------------------------|
| clean_workforce_dataset.xlsx     | Cleaned dataset used by the implementation.        |
| diversity_analytics_outputs.xlsx | Dashboard-ready diversity summary tables.          |
| diversity_analytics.py           | Reusable Python implementation.                    |
| charts/                          | Generated diversity and comparison visualizations. |
| README.md                        | Implementation summary and dataset status.         |

All outputs listed above are included in the Milestone 3 clean implementation ZIP.

## 12. Implementation Logic

- Read the uploaded Excel dataset.
- Clean text values and remove exact duplicate rows.
- Validate the required diversity fields.
- Create age groups.
- Calculate gender, age and demographic summaries.
- Calculate department-level gender comparisons.
- Calculate department-level age statistics.
- Generate charts using the cleaned data.
- Export dashboard-ready Excel sheets.
- Store all implementation outputs in the project package.

## 13. Final Milestone 3 Output

The completed Clean Diversity Analytics Module satisfies the assigned Milestone 3 scope by analyzing workforce gender, age and demographic metrics and producing charts and comparison visualizations.

**Final Workflow:** Clean Workforce Dataset → Gender Analysis + Age Analysis + Demographic Analysis → Comparison Visualizations → Dashboard-Ready Excel Outputs

The implementation is ready to be used as the diversity-analysis component of the interactive workforce intelligence dashboard.